



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A' Grade

Department of Computer Science and Engineering DevOps Lab - L3

Lab Assignment No: 12	
Name of Student	Aryan Sheladia
PRN	23070122202
Batch	2023 - 2027
Class	CSE – B1
Academic Year & Semester	2026 - 2027, 7th Semester
Title of Assignment	Kubernetes Pod & Deployment
Implementation & Theory	<pre>PS E:\DevOps_SEM_7> kubectl cluster-info Kubernetes control plane is running at https://127.0.0.1:62602 CoreDNS is running at https://127.0.0.1:62602/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'. PS E:\DevOps_SEM_7> kubectl get nodes NAME STATUS ROLES AGE VERSION minikube Ready control-plane 5m38s v1.37.0 PS E:\DevOps_SEM_7></pre> Local Kubernetes Cluster Setup: Minikube was started using the Docker driver to create a local Kubernetes cluster. The cluster was verified using <code>kubectl cluster-info</code> and <code>kubectl get nodes</code>, confirming that the Minikube node was successfully running and ready. <pre>PS E:\DevOps_SEM_7> docker images my-flask-app REPOSITORY TAG IMAGE ID CREATED SIZE my-flask-app v2 e647443a25d2 13 days ago 200MB my-flask-app latest 2aec46790c9e 2 weeks ago 200MB PS E:\DevOps_SEM_7> minikube image load my-flask-app:v2 PS E:\DevOps_SEM_7> minikube image ls Select-String "my-flask-app" docker.io/library/my-flask-app:v2 PS E:\DevOps_SEM_7></pre> Image Accessibility: The locally built <code>my-flask-app:v2</code> Docker image was loaded into the Minikube cluster so that Kubernetes could access the image locally without downloading it from an external container registry.

```
web-deployment.yaml U X
Assignment - 12 > web-deployment.yaml
1  apiVersion: apps/v1
2  kind: Deployment
3
4  metadata:
5    name: web-deployment
6
7  spec:
8    replicas: 2
9
10   selector:
11     matchLabels:
12       app: web-app
13
14   template:
15     metadata:
16       labels:
17         app: web-app
18
19     spec:
20       containers:
21         - name: web
22           image: my-flask-app:v2
23           imagePullPolicy: IfNotPresent
24           ports:
25             - containerPort: 5000
```

Kubernetes Deployment Configuration: A Kubernetes Deployment manifest was created to deploy the Flask application using the my-flask-app:v2 image. The configuration specifies two replicas, ensuring that Kubernetes creates and maintains two identical application Pods. The label app: web-app is used to identify and manage the Pods.

```
PS E:\DevOps_SEM_7\Assignment - 12> kubectl apply -f web-deployment.yaml
deployment.apps/web-deployment created
PS E:\DevOps_SEM_7\Assignment - 12> kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
web-deployment      2/2     2            2           14s
PS E:\DevOps_SEM_7\Assignment - 12> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
web-deployment-7f75cdd746-khwln     1/1     Running   0           21s
web-deployment-7f75cdd746-mc6dr     1/1     Running   0           21s
PS E:\DevOps_SEM_7\Assignment - 12>
```

Deployment Creation: The Deployment was applied to the Minikube Kubernetes cluster using kubectl apply. Kubernetes created and managed two replicas of the Flask application, and the Pods reached the Running state, confirming successful deployment.

```
PS E:\DevOps_SEM_7\Assignment - 12> kubectl get pods -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
web-deployment-7f75cdd746-khw1n	1/1	Running	0	81s	10.244.0.3	minikube	<none>	<none>
web-deployment-7f75cdd746-mc6dr	1/1	Running	0	81s	10.244.0.4	minikube	<none>	<none>

Pod Verification: The detailed Pod status confirms that two replicas of the Flask application are running successfully. Kubernetes automatically created and manages these Pods through the Deployment controller.

```
PS E:\DevOps_SEM_7\Assignment - 12> kubectl describe deployment web-deployment
```

```
Name: web-deployment
Namespace: default
CreationTimestamp: Fri, 04 Sep 2026 03:04:33 +0530
Labels: <none>
Annotations: deployment.kubernetes.io/revision: 1
Selector: app=web-app
Replicas: 2 desired | 2 updated | 2 total | 2 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
```

Deployment Management: The Deployment controller maintains the desired state of two application replicas. Kubernetes monitors the Pods and ensures that the required number of replicas remains available.